CSE 216 Formal Methods
Fall 2026 Daniel Fremont

HW 3

Due October 20 at 3:00 pm

(4 questions, 185 points total)

I discussed questions with vadim. specifically 2c and 1d

1. (20 pts.) Resolution

(a) Which of the following clauses can be resolved with each other, and what are the resulting clauses?

(A) $x \vee y \vee z$
(B) y
(C) $\neg y$
(D) $\neg y \vee z$

Answer:

(a) clauses A,C, results in resolution of $x \vee z$
(b) clauses A,D results in $x \vee z$
(c) clauses B,C results in $\bot$
(d) clauses B,D results in z

(b) Give a resolution refutation for the UNSAT formula with the following clauses:

$$c_1 = \neg x \vee \neg y \vee z$$
$$c_2 = \neg x \vee w$$
$$c_3 = x \vee \neg y$$
$$c_4 = \neg y \vee \neg z \vee \neg w$$
$$c_5 = x \vee y$$
$$c_6 = y \vee \neg w$$

You may write $c_i \odot c_j$ to denote the resolution of two clauses.

Answer:

i. $c_7 = c_3 \odot c_5 = (x \vee \neg y) \odot (x \vee y) = x$
ii. $c_8 = c_2 \odot c_7 = (\neg x \vee w) \odot (x) = w$
iii. $c_9 = c_6 \odot c_8 = (y \vee \neg w) \odot (w) = y$
iv. $c_{10} = c_1 \odot c_7 = (\neg x \vee \neg y \vee z) \odot (x) = \neg y \vee z$
v. $c_{11} = c_4 \odot c_9 = (\neg y \vee \neg z \vee \neg w) \odot (y) = \neg z \vee \neg w$
vi. $c_{12} = c_{11} \odot c_8 = (\neg z \vee \neg w) \odot (w) = \neg z$
vii. $c_{13} = c_1 \odot c_{12} = (\neg x \vee \neg y \vee z) \odot (\neg z) = \neg x \vee \neg y$
viii. $c_{14} = c_{13} \odot c_7 = (\neg x \vee \neg y) \odot (x) = \neg y$
ix. $c_{15} = c_{14} \odot c_9 = (\neg y) \odot (y) = \bot$

2. (60 pts.) CDCL

Consider running CDCL on the formula in (1b) above, with a variable selection heuristic that prefers $z > w > x > y$ and a phase selection heuristic that always picks the negative polarity.

(a) Up to the first conflict, what decisions are made?

Answer: z is false at DL1

w=false at DL2

x is false from c_2 and then y is false from c_3. this leads to c_5 being unsat.

(b) Construct the implication graph for the first conflict.

Answer: see figure 1

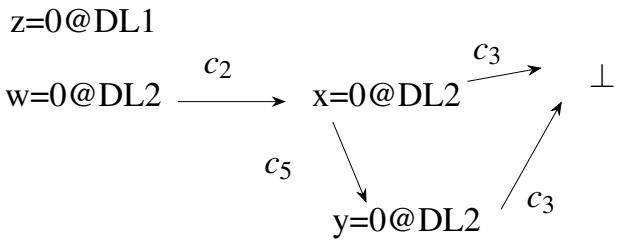


Figure 1: 2b

(c) What are the possible conflict clauses that could be learned from this conflict? Which of these is asserting if we backtrack one level?

Answer: we could learn

(a) $x \vee y$

(b) x

(c) w

we can say that w must be true if we go back to DL1.

(d) If we use the first UIP heuristic, which conflict clause will CDCL learn, and what decision level will it backtrack to?

Answer: the conflict clause learned is x. it will backtrack to DL0 since the "second highest" is the empty clause which is at DL0.

(e) Construct the implication graph for the next conflict.

Answer: see fig 2

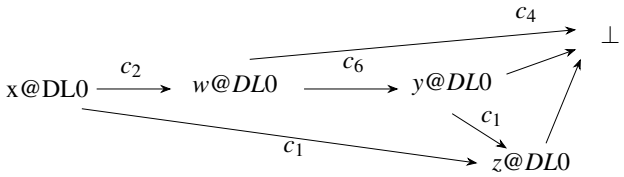


Figure 2: 2e

(f) CDCL terminates after these two conflicts. Why? How can you use the implication graphs from each conflict to build a resolution refutation of the original formula?

Answer: the second conflict happens at DL0 which means the formula is unsat.

$c_3(x \vee y)$ resolved with $c_5(x \vee y)$ on y gives x.

c_2, c_6, c_1, c_4 with x resolve to $\perp$

3. (70 pts.) First-order logic

For each of the following formulas, state whether they are satisfiable or valid (or neither). If a formula ϕ is satisfiable, give a model, i.e. a structure M such that $M \models \phi$. If ϕ is not valid, give a structure M such that $M \not\models \phi$. For each formula, assume the signature of the logic — the constant, function, and relation symbols — consists just of those symbols used in the formula. We use c, f, and R as constant, function, and relation symbols respectively.

(a) $\forall x.\forall y.(f(x,y) = f(y,x))$

Answer: sat but not valid

if f is multiplication and x and y are integers, the statement is $\top$

if f is multiplication and x and y are integer matricies, most matricies do not commute and the statement is $\bot$

(b) $\exists x.(x = x)$ *(hint: can you pick the universe of the structure so that this is false?)*

Answer: sat but not valid

any universe with more than a single object, x=x and therefore sat

any universe with no objects, there are no x's to satisfy the equation and therefore not valid

(c) $\forall x.\forall y.(x = y)$

Answer: sat but not valid

any universe with one object the formula is true.

any universe with more than 1 object, say natural numbers, not all x equals y.

(d) $\exists x.(x = x) \wedge \forall x.\forall y.\neg(x = y)$

Answer: assuming that you can assume different domains for the $\exists x$ on the left and the $\forall x.\forall y$ on the right, sat but not valid

the LHS is invalid and so the whole expression is invalid by the prior question

this is sat, let x in the lhs of the and be natural numbers and let x and y on the right side be elements in the empty set. the left is true (ex: 1=1), and the right is vacuously true.

(e) $\forall x.\exists y.(f(x) = y)$

Answer: sat but not valid if y is not of the correct "sort", as in f sends elements of x to elements of "sort" $sort(y)$, then this is true.

ex: x and y are nat and f is the successor function.

it could be not valid if x is nat, f is succ, and y a string.

(f) $\exists x.(R(x) \wedge \neg R(f(x)))$

Answer: Sat but not valid

proof of non validity:

x is a variable in the set $\{n | n \in \mathbb{N}^+ \wedge n < 2\}$. let R be the inclusion of natural numbers onto the integers, then output true if it is less than 0. let f be the identity function. this formula is $\bot$.

proof of sat: x is the same as validity example. let f be the inclusion map onto integers then multiply by -1. let R accept integer and natural numbers and output true if x is greater than 0. x=1 satisfies the formula.

(g) $[R(c) \wedge \forall x.(R(x) \rightarrow R(f(x)))] \rightarrow R(f(f(f(c))))$

Answer: this answer is sat and valid.

sat: let the domain be $\{1\}$, let f be the identity function and $R(x)$ be true if it is the case that $x = 1$.

4. (35 pts.) Arrays

Consider the theory of bitvectors (say, 32-bit) and arrays whose indices and values are bitvectors (sometimes called ABV). This theory has two kinds of objects, or *sorts*: bitvectors and arrays. Let's use the notation $\forall x : Array.$ and $\forall x : BV.$ to quantify over arrays and bitvectors respectively. So for example if we use the C notation $>>$ for the right shift operator, then $\forall x : BV.(x >> 32 = 0)$ would be a valid formula asserting that for all 32-bit bitvectors, right shifting by 32 bits yields the zero bitvector.

(a) Write an SMT formula $\phi(x)$ which is true if and only if the array x is sorted in increasing order.
Answer: $\forall(y : BV).\forall(z : BV).((y \leq z) \rightarrow (select(x,y) \leq select(x,z)))$

(b) Write a formula $\phi(x)$ which asserts that every element of the array x has the same value.
Answer: $\exists(z : BV).\forall(y : BV).(select(x,y) = z)$

(c) Write a formula with no free variables (a *sentence*) asserting that if element 0 of an array is even, and for every i if element i is even then element $i+1$ is even, then all elements of the array are even. Is this formula valid?
Answer:

$$\begin{aligned}&\forall c : Array.\\&[((\exists(a : BV).select(c,0) = 2 \cdot a) \wedge\\&(\forall(x : BV).(\exists(a : BV).select(c,x) = 2a \rightarrow \exists(b : BV).select(c,x+1) = 2b))) \rightarrow\\&\forall(x : BV).\exists(a : BV).select(c,x) = 2a]\end{aligned} \tag{1}$$

The formula should be valid. the wrap around doesnt affect the truth.

5. (optional, bonus problem; 0 pts.) GCHQ puzzle

The British signals intelligence organization GCHQ released a puzzle in December 2015, the correct solution of which led to a sequence of increasingly difficult puzzles (presumably with your employment as a codebreaker at the end — alas, the puzzles have been taken down). They described it as "a brain-teasing puzzle that seems certain to exercise the grey matter of participants over the holiday season". Don't exert yourself: solve it with SAT.